

Mostafa Abdallah

Data Scientist | AI Engineer | Full-Stack Developer Cairo, Egypt

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PROFESSIONAL SUMMARY

Results-driven Software & AI Engineer with expertise in building scalable machine learning models, computer vision systems, and full-stack data pipelines. Proven track record of delivering end-to-end AI solutions from advanced healthcare diagnostics to real-time edge detection systems ready to drive impactful, data-driven innovations.

EDUCATION

The Egyptian Chinese University (ECU) *B.Sc. Engineering, Software Engineering* | Aug 2021 – Jul 2026

- **Grade:** Very Good
- **Honors:** 2nd Place National Winner in Micromouse Robotics Competition (C++ Autonomous Navigation).
- **Activities:** ACM ICPC (International Collegiate Programming Contest) Contestant.
- **Coursework:** Artificial Intelligence, Machine Learning, Computer Vision, Data Structures, OOP.

EXPERIENCE

Data Science & AI Trainee (Data Scientist Track) *Digital Egypt Pioneers Initiative (DEPI) - IBM* | Jun 2025 – Jan 2026

- Advanced expertise in AI and Data Science through an intensive IBM-backed diploma and training program.
- Worked extensively on real-world datasets, deploying machine learning algorithms, database management, and Python-based data pipelines.
- **Skills:** Data Science, Machine Learning, Python, Database Management, Data Analysis.

PROJECTS

Aethea: AI-Powered Healthcare & Medical Image Diagnostics Platform

- Engineered a centralized healthcare platform connecting patients and doctors, featuring automated medical scan analysis (Fracture detection) using AI models with GradCAM explainability.
- Architected the full-stack system utilizing React 19, Tailwind, and Node.js, backed by a Supabase (PostgreSQL) and Redis infrastructure.
- **Skills:** Deep Learning, GradCAM, React 19, Node.js, Supabase, Redis, Full-Stack Architecture.

VisionGuard AI: Real-Time Exam Cheating Detection System

- Developed an automated AI-driven desktop application (using CustomTkinter) to monitor examination integrity and flag suspicious student behavior in real-time.
- Integrated a custom YOLOv10 model with Fast-Motion ByteTrack for object tracking, automating evidence collection with seamless cloud uploading to Firebase.
- **Skills:** YOLOv10, ByteTrack, Computer Vision, Firebase, CustomTkinter, Python.

PricePulse Airbnb (Data Engineering & Dashboards)

- Engineered a complete Data Engineering pipeline to scrape, clean, and analyze real Airbnb market data across 10 European cities using Selenium.
- Implemented a Medallion Architecture (Bronze, Silver, Gold) using MySQL, and designed interactive Power BI dashboards to extract actionable business insights on pricing.
- **Skills:** Data Engineering, Web Scraping (Selenium), MySQL, Power BI, Data Visualization.

Smart Home Energy Forecaster

- Developed an end-to-end Deep Learning pipeline using 1D CNNs to forecast smart home energy consumption and classify weather conditions.
- Implemented rigorous data preprocessing and actively tracked computational complexity (FLOPs) and carbon emissions during training to optimize environmental footprint.
- **Skills:** Deep Learning, 1D CNN, TensorFlow/Keras, Time-Series Analysis, Feature Engineering.

NLP & Classification Systems: Complaint, Emotion & NER

- Engineered multiple NLP pipelines utilizing RoBERTa and BiLSTM-CRF to categorize financial complaints (using PyTorch FP16 on dual GPUs), detect emotional sentiment, and extract complex named entities.
- Deployed the winning classification models into fully interactive Gradio and Streamlit dashboards for live sentiment analysis and user feedback evaluation.
- **Skills:** NLP, Text Classification, RoBERTa, PyTorch (FP16), Transformers, Gradio, Streamlit.

CERTIFICATIONS & TRAINING

AI, Data Science & Machine Vision Diploma | *Instant* | Jun 2025 – Present

- Completing a comprehensive diploma focused on building end-to-end data pipelines, interactive dashboards, and training advanced computer vision architectures.

Full-Stack Web Development (Frontend & Backend) | *SEF Academy* | May 2024 – Feb 2025

- Developed full-stack web applications by building responsive user interfaces and integrating them with secure, scalable backend architectures (MERN Stack).

CCNA (Cisco Certified Network Associate) | *Jelecom* | Jan 2023 – May 2023

- Gained practical knowledge in enterprise networking, configuring routers/switches, and network security.

TECHNICAL SKILLS

- **AI & Machine Learning:** PyTorch, TensorFlow, Scikit-learn, Transformers, OpenCV, YOLO, NLTK, Spacy.
- **Programming Languages:** Python, C++, JavaScript, TypeScript, Java, SQL.
- **Web, MLOps & Tools:** React.js, Node.js, Supabase, Gradio, Streamlit, Firebase, Docker, Git, Power BI.
- **Languages:** Arabic (Native), English (Professional Working Proficiency)